

Exp3

The screenshot shows a Wireshark packet capture of a DNS response. The packet list pane on the left shows a sequence of packets, with packet 86 selected. The packet details pane on the right shows the structure of the selected packet, which is a DNS response from 172.16.162.88 to 8.8.8.8. The packet is a standard query response for 0x7a13 A detectportal.firefox.com CNAME mozilla.map.fastly.net A. The packet is 195 bytes long and is a standard query response. The packet is a standard query response for 0x7a13 A detectportal.firefox.com CNAME mozilla.map.fastly.net A. The packet is a standard query response for 0x7a13 A detectportal.firefox.com CNAME mozilla.map.fastly.net A.

No.	Time	Source	Destination	Protocol	Length	Info
82	26.697408723	172.16.162.88	8.8.8.8	DNS	95	Standard query 0x7a13 A detectportal.firefox.com OPT
84	26.697676118	172.16.162.88	8.8.8.8	DNS	95	Standard query 0x3a66 AAAA detectportal.firefox.com OPT
85	26.714214156	8.8.8.8	172.16.162.88	DNS	243	Standard query response 0x3a66 AAAA detectportal.firefox.com CNAME mozilla.map.fastly.net A
86	26.726153950	8.8.8.8	172.16.162.88	DNS	195	Standard query response 0x7a13 A detectportal.firefox.com CNAME mozilla.map.fastly.net A
97	26.750270987	172.16.162.88	8.8.8.8	DNS	82	Standard query 0xebf6 A example.org OPT
98	26.750598905	172.16.162.88	8.8.8.8	DNS	82	Standard query 0x7125 AAAA example.org OPT
99	26.751089083	172.16.162.88	8.8.8.8	DNS	84	Standard query 0x416f AAAA ipv4only.arpa OPT
101	26.765611623	8.8.8.8	172.16.162.88	DNS	114	Standard query response 0xebf6 A example.org A 104.20.26.136 A 172.66.157.237 OPT
102	26.768855420	8.8.8.8	172.16.162.88	DNS	141	Standard query response 0x416f AAAA ipv4only.arpa SOA sns.dns.icann.org OPT
108	26.787730691	8.8.8.8	172.16.162.88	DNS	138	Standard query response 0x7125 AAAA example.org AAAA 2606:4700:10::ac42:9ded AAAA 2606:4...

Flags: 0x00
0... .. = Reserved bit: Not set
.0... .. = Don't fragment: Not set
..0... .. = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 122
Protocol: UDP (17)
Header Checksum: 0x0e3c [validation disabled]
[Header checksum status: Unverified]
Source Address: 8.8.8.8
Destination Address: 172.16.162.88

User Datagram Protocol, Src Port: 53, Dst Port: 49333
Source Port: 53
Destination Port: 49333
Length: 161
Checksum: 0x09f7 [unverified]
[Checksum Status: Unverified]
[Stream index: 18]
[Timestamps]
[Time since first frame: 0.022745227 seconds]
[Time since previous frame: 0.022745227 seconds]
UDP payload (153 bytes)

Domain Name System (response)
Transaction ID: 0x7a13
Flags: 0x8180 Standard query response, No error
1... .. = Response: Message is a response
.000 0... .. = Opcode: Standard query (0)

The screenshot shows a Wireshark packet capture of a DNS response. The packet list pane on the left shows a sequence of packets, with packet 86 selected. The packet details pane on the right shows the structure of the selected packet, which is a DNS response from 172.16.162.88 to 8.8.8.8. The packet is a standard query response for 0x7a13 A detectportal.firefox.com CNAME mozilla.map.fastly.net A. The packet is 195 bytes long and is a standard query response. The packet is a standard query response for 0x7a13 A detectportal.firefox.com CNAME mozilla.map.fastly.net A.

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Flags: 0x00
0... .. = Reserved bit: Not set
.0... .. = Don't fragment: Not set
..0... .. = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 122
Protocol: UDP (17)
Header Checksum: 0x0e3c [validation disabled]
[Header checksum status: Unverified]
Source Address: 8.8.8.8
Destination Address: 172.16.162.88

User Datagram Protocol, Src Port: 53, Dst Port: 49333
Source Port: 53
Destination Port: 49333
Length: 161
Checksum: 0x09f7 [unverified]
[Checksum Status: Unverified]
[Stream index: 18]
[Timestamps]
[Time since first frame: 0.022745227 seconds]
[Time since previous frame: 0.022745227 seconds]
UDP payload (153 bytes)

Domain Name System (response)
Transaction ID: 0x7a13
Flags: 0x8180 Standard query response, No error
1... .. = Response: Message is a response
.000 0... .. = Opcode: Standard query (0)

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dns

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[Timestamps]
[Time since first frame: 0.022745227 seconds]
[Time since previous frame: 0.022745227 seconds]
UDP payload (153 bytes)

Domain Name System (response)
Transaction ID: 0x7a13

Flags: 0x8100 Standard query response, No error

1... .. = Response: Message is a response
.000 0... .. = Opcode: Standard query (0)
... ..0... .. = Authoritative: Server is not an authority for domain
... ..0... .. = Truncated: Message is not truncated
... ..1... .. = Recursion desired: Do query recursively
... ..1... .. = Recursion available: Server can do recursive queries
... ..0... .. = Z: reserved (0)
... ..0... .. = Answer authenticated: Answer/authority portion was not authenticated by the server
... ..0... .. = Non-authenticated data: Unacceptable
... ..0000 = Reply code: No error (0)

Questions: 1
Answer RRs: 5
Authority RRs: 0
Additional RRs: 1

Queries
detectportal.firefox.com: type A, class IN

Answers
detectportal.firefox.com: type CNAME, class IN, cname mozilla.map.fastly.net
mozilla.map.fastly.net: type A, class IN, addr 151.101.129.91
mozilla.map.fastly.net: type A, class IN, addr 151.101.65.91

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Destination Port: 53
Length: 61
Checksum: 0x5ec7 [unverified]
[Checksum Status: Unverified]
[Stream index: 18]

[Timestamps]
[Time since first frame: 0.000000000 seconds]
[Time since previous frame: 0.000000000 seconds]
UDP payload (53 bytes)

Domain Name System (query)
Transaction ID: 0x7a13

Flags: 0x0100 Standard query

0... .. = Response: Message is a query
.000 0... .. = Opcode: Standard query (0)
... ..0... .. = Truncated: Message is not truncated
... ..1... .. = Recursion desired: Do query recursively
... ..0... .. = Z: reserved (0)
... ..0... .. = Non-authenticated data: Unacceptable

Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1

Queries
detectportal.firefox.com: type A, class IN

Additional records
<Root>: type OPT
[Response in: 86]